Signed order count and order-book imbalance

TSLA, 2019

Conditional impact, chronological prediction, and depth-aware replay

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This analysis began in the course project *Liquidity Games in High-Frequency Markets*, supervised by Anastasia Bugaenko of Capital Fund Management. The course presentation was prepared by L. Cohen, R. Darwish, A. Pariente, H. Rohrbach, and R. Sithisak. I later rewrote the supervised analysis and prepared this report in a separate repository.

Abstract

The analysis reconstructs 4,547,078 visible market orders from 249 included sessions among 252 delivered TSLA file pairs in 2019. LOBSTER messages are aligned with the book state immediately before each event, and fills with the same timestamp and initiating side are grouped. The main experiment relates signed volume and signed order count in a 10-order window to the mid-price change over that window. Since the regressors are observed during the window, the exercise estimates conditional impact rather than a pre-trade forecast.

The fixed late-year dates form an untrimmed holdout of 102,468 windows. Signed-volume OLS reaches $R^2 = 0.118$. Nonlinear volume terms raise the score to 0.298, and raw signed order count raises it to 0.359. Further count transforms change little: the full score is 0.360. Adding raw count to the nonlinear volume model reduces mean squared error by 8.7%. Resampling complete test dates gives a 95 percent interval of 8.0% to 9.5% for that incremental reduction. The full model's total reduction relative to linear signed volume is 27.4%.

Separately, a C++20 engine audits 38,516,432 included level-2 message/snapshot pairs without claiming full order priority. It classifies 28,544,808 transitions as exact and 9,971,375 as depth-censored, with no observable mismatch, unsupported event, or invalid snapshot. A forward-looking experiment then predicts the direction of the next same-session mid-price change from current queue imbalance and causal top-of-book OFI. The combined model reaches ROC-AUC 0.627 and reduces Brier score by 5.61% across all spreads. In the exploratory one-tick subset, the corresponding values are 0.752 and 18.24%.

I also apply the finite-size scaling construction of Patzelt and Bouchaud (2018) to the same stock-year. The width and height scale regressions have in-sample R^2 values of 0.9958 and 0.9975. These fits use 28 estimated scale points and are unrelated to the chronological prediction score.

Source-data status. The original project access has ended and replacement files are unavailable. A shared policy declares the 3 delivered files that end before their scheduled close as source exclusions. The Python gate audits all 252 pairs, and the C++ validator applies the same rules to decoded sessions. The shared counts require exactly 249 included sessions with no undeclared coverage failure. Every reported artifact was regenerated on that universe; the study does not claim complete 2019 coverage or external replication.

Keywords: market impact, queue imbalance, order-flow imbalance, market microstructure, chronological validation, order-book replay, finite-size scaling

1 Question

In the local benchmark of Kyle (1985), signed order flow and price change are connected by a linear slope. Empirical impact curves bend, change scale with the aggregation horizon, and reflect persistent trade signs (Bouchaud et al., 2018; Patzelt and Bouchaud, 2018). I ask two distinct questions for TSLA in 2019. First, does signed order count explain conditional price impact on later dates after signed volume has already been included? Second, do the current queues and order-flow changes observed since the last price move forecast the direction of the next move?

Signed volume alone explains 0.118 of the holdout variance. Fixed volume transforms explain 0.298. Adding raw signed count brings the score to 0.359, almost the same as the full specification at 0.360. Raw signed count provides the second, incremental gain after volume curvature. Extra transforms and ridge regression leave the reported score unchanged.

Both regressors and response cover the same event-time window. The date split tests whether a conditional mapping estimated earlier in the year transfers to later dates. It does not identify a causal effect because order flow and liquidity react to each other.

The primary holdout keeps every observation. I estimate uncertainty by resampling trading dates, then repeat the comparison over expanding test blocks, five window lengths, and a basis-point response. The earlier trimmed protocol is reported separately because it produced the original 0.24 to 0.38 comparison.

2 Relation to prior work

Kyle's model gives a local liquidity measure: a larger price response per unit of signed flow corresponds to a less deep market (Kyle, 1985). It does not imply that the empirical response must remain linear far from balanced flow. The evidence reviewed by Bouchaud et al. (2018) shows why impact is difficult to see at the trade level and why conditional averaging is central to measurement.

Patzelt and Bouchaud (2018) study volume and sign imbalance over several intraday horizons. Their conditional curves can be rescaled by a horizon-dependent width and height. The published claim is based on many instruments and periods. Here, the same functional form is fitted to one included TSLA stock-year. This within-sample application can assess its description of these sessions; it is not an external replication of the paper's cross-sectional evidence.

The course project followed Bugaenko (2019), which connects order-flow imbalance, local liquidity estimates, and supervised impact models. I retain the transaction reconstruction and scaling exercise, but rebuild the supervised evaluation around untrimmed chronological date splits. The model ablation isolates signed count as the useful extension of a nonlinear volume baseline.

Gould and Bonart (2015) study queue imbalance as a one-tick-ahead price predictor and emphasize its dependence on the spread regime. I use the displayed level-1 imbalance for a separate next-mid-price-direction experiment. The pooled sample is the primary comparison; the one-tick subset is reported as exploratory.

3 Data and transaction reconstruction

3.1 LOBSTER alignment

LOBSTER provides a message file and an order-book file for each session. Message row k is the event that moves the book from row $k-1$ to row k (LOBSTER, 2026). The pre-event best bid and ask for an execution on row k therefore come from order-book row $k-1$. The pipeline makes that shift before calculating the mid-price or spread. An execution on the first row of a session is discarded because its pre-event state is absent.

Prices in the source files are integers scaled by 10,000. For a visible execution, event type 4, LOBSTER's direction field identifies the resting order. The aggressor sign is its opposite: $+1$ for a buyer-initiated execution and -1 for a seller-initiated execution.

3.2 Source coverage

LOBSTER filenames state the requested time range, but that range alone does not show that the delivered rows reach the close. The audit therefore finds the last message at or before the scheduled close and requires it to fall within 60 seconds. Nasdaq's 2019 notices set a 13:00 close on 3 July, 29 November, and 24 December; the other files are checked against 16:00.

249 of the 252 delivered sessions pass. The files for 9 January, 8 March, and 18 September end 22,827, 958, and 5,403 seconds early. The original project access has ended and replacement files are unavailable, so the shared analysis policy declares exactly those three dates as source exclusions. Both Python annual readers audit every delivered pair, include the other 249 sessions, and reject any additional incomplete input before writing annual output. The C++ validator consumes the same policy for decoded sessions and exact universe counts. The daily aggregate record is committed as `results/session_coverage.csv`; it contains no licensed row, event price, or order identifier. Rows after the scheduled close are counted in the audit and excluded from prepared sessions: 909 on 3 July, 301 on 29 November, and 774 on 24 December. The fixed policy preserves development through 6 August, selection from 7 August through 17 October, and test from 18 October, giving 148/50/51 nested dates and a 198/51 outer split.

A market order may consume several displayed orders. The source used here does not expose a parent market-order identifier, so visible fills with the same date, timestamp, and aggressor sign are grouped. Share size is summed, execution price is volume weighted, and the first pre-event book defines the starting state. This is a reconstruction rule, not investor identification.

3.3 Sample

Table 1 reports descriptive quantities, not model results. The median reconstructed order has 55 shares. The distribution is right-skewed: its 90th and 99th percentiles are 195 and 700 shares. The median quoted spread is 7.0 cents, and the share of buyer-initiated orders is 49.7%.

Table 1. Reconstructed visible-order sample. Daily quantiles are computed across 249 sessions; order-level quantiles use all reconstructed visible market orders.

Quantity	Value
Timestamp-aggregated visible market orders	4,547,078
Median orders per session	16,049
10th to 90th percentile, orders per session	10,839 to 29,935
Median order size, shares	55
90th and 99th percentile order size, shares	195, 700
Median and 90th percentile spread, cents	7.0, 15.0
Buyer-initiated fraction	49.7%
Non-overlapping 10-order windows	454,576
Standard deviation of 10-order impact, cents	12.46
Zero-impact share of 10-order windows	3.0%

The raw and derived row-level files remain outside the repository because the LOBSTER license does not permit redistribution. Committed CSV and JSON files contain aggregate curves, scores, and sample summaries only.

3.4 A separate scaling table

The supervised experiment uses visible executions because their aggressor side is given by the resting-order direction. The scaling exercise also uses hidden executions, event type 5, to match the broader transaction definition in Patzelt and Bouchaud (2018). Their signs are inferred from execution price relative to the pre-event mid-price. Midpoint executions have no inferred side and are removed. After timestamp grouping, this table contains 5,845,437 transactions. The prediction and scaling exercises therefore use different transaction tables.

4 Measurement

Consider a same-session window beginning at reconstructed market order t and containing T signed orders. With aggressor sign ϵ_i, share size v_i, and pre-event mid-price m_i, define

$$\Delta V_{t,T} = \sum_{i=t}^{t+T-1} \epsilon_i v_i, \tag{1}$$

$$\Delta E_{t,T} = \sum_{i=t}^{t+T-1} \epsilon_i, \tag{2}$$

$$Y_{t,T} = 100\,(m_{t+T} - m_t)\,. \tag{3}$$

Here, Y is measured in cents. Windows are non-overlapping within each horizon and never cross a session boundary. A T-order window needs the mid-price at its right edge, so the final incomplete block of each session is discarded.

The count imbalance ΔE distinguishes flow patterns that signed volume alone merges. One large buy and several smaller buys can generate comparable ΔV but different ΔE. That difference may reflect order splitting, replenishment, or state-dependent order size. The regression cannot separate those mechanisms, but it can test whether the count adds conditional information on dates not used for estimation.

Figure 1 displays the raw conditional mean at four horizons. Quantile bins are used only for the figure. The outer bins contain wide ranges of volume, while the central bins reveal a steep response around balanced flow. The flattening in both tails is visible at every horizon. A single global linear coefficient cannot capture this shape.

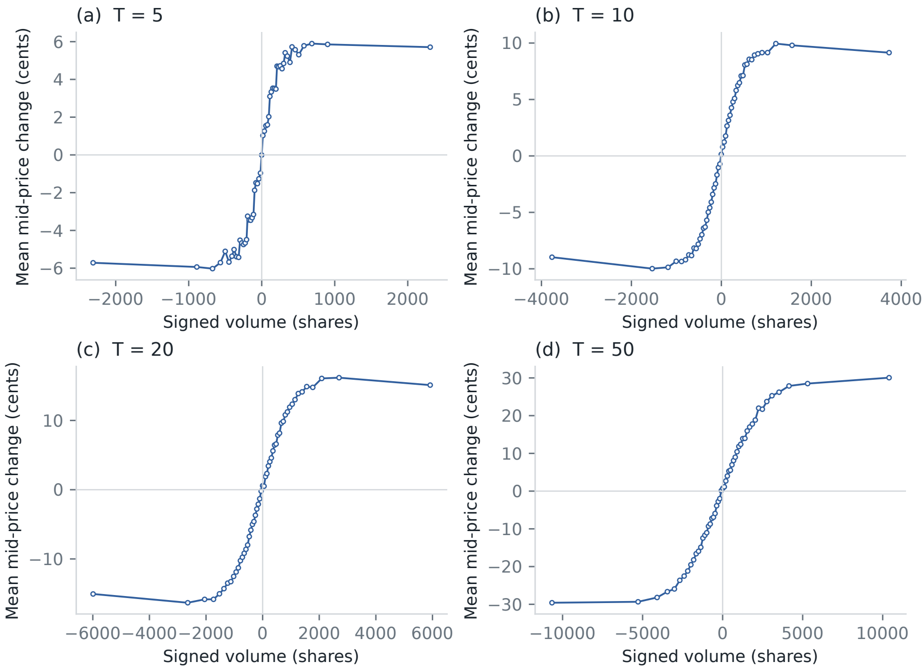

Figure 1. Mean mid-price change conditional on signed volume. Each point is a mean within one of 51 signed-volume quantiles. All 249 dates enter; the bins are not used for model fitting.

A local Kyle slope is also estimated within the central 35 percent of absolute signed volume. At $T = 10$, the fitted value is 0.01837 cents per share. Across $T \in \{5, 10, 20, 50, 100\}$, the fitted slopes decay approximately as $T^{-0.315}$. This is a descriptive liquidity estimate, not a trade-level execution-cost model.

5 Evaluation design

5.1 Chronological holdout

The main $T = 10$ sample contains 454,576 windows across 249 dates. Dates through 17 October form the training partition and dates from 18 October form the holdout. This fixed calendar boundary was retained after the source exclusions, so no inspected test date moved into training. It contains 198 included training dates and 51 holdout dates. Table 2 gives the exact boundary. No row is removed from either side of the split.

Table 2. Primary chronological partition.

Partition	First date	Last date	Windows
Train	2019-01-02	2019-10-17	352,108
Holdout	2019-10-18	2019-12-31	102,468

The four nested OLS specifications are:

$$
\begin{aligned}
&\mathcal{M}_1 : \Delta V, \\
&\mathcal{M}_2 : \Delta V, |\Delta V|, \operatorname{sgn}(\Delta V)\sqrt{|\Delta V|}, \operatorname{sgn}(\Delta V)\log(1+|\Delta V|), \\
&\mathcal{M}_3 : \mathcal{M}_2, \Delta E, \\
&\mathcal{M}_4 : \mathcal{M}_3, |\Delta E|, \operatorname{sgn}(\Delta E)\sqrt{|\Delta E|}.
\end{aligned}
$$

The transforms are fixed before estimation. An intercept is included. Ridge regression with standardized $\mathcal{M}_4$ features is retained as a numerical check, but its predictions are indistinguishable from OLS at the shown precision.

Scores are calculated only on the holdout:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2, \tag{4}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{5}$$

Mean absolute error and sign accuracy are reported as secondary diagnostics. Sign accuracy omits a window when either the realized or fitted response is exactly zero.

5.2 Uncertainty and stability checks

Event-time windows from one session share market conditions. Treating all windows as independent would understate uncertainty. The bootstrap therefore resamples the 51 complete holdout dates with replacement. For each of 10,000 repetitions, scores are recomputed from the selected daily blocks. The interval is the 2.5th to 97.5th percentile of the bootstrap distribution.

Five additional temporal splits begin with 40 percent of the dates for training. Each subsequent block is tested once, then added to the expanding training set. Horizon checks retain the 18 October boundary for $T = 5, 10, 20, 50, 100$. Finally, the response is replaced by $10{,}000\log(m_{t+T}/m_t)$ to check that the result is not an artifact of measuring a changing stock price in cents.

6 Holdout results

6.1 Feature ablation

Table 3 separates curvature from count information. The linear volume baseline has $R^2 = 0.118$. Fixed volume transforms account for the first increase, to 0.298. Adding only raw signed count reaches 0.359. The two additional count transforms change the third decimal place by less than one point. Raw signed count accounts for the useful part of the second increase.

Table 3. Full chronological holdout, $T = 10$. Errors are measured in cents.

Specification	R^2	MSE	MAE	Sign accuracy
Signed volume	0.118	147.23	8.64	77.0%
Volume transforms	0.298	117.16	7.79	76.9%
Volume transforms plus raw signed count	0.359	106.93	7.48	78.5%
Volume and count transforms	0.360	106.86	7.47	78.5%

Figure 2 compares fitted and realized means in 20 holdout bins. The nonlinear volume-only model still leaves a systematic count residual. Adding raw signed count follows the binned means more closely, although individual windows remain noisy.

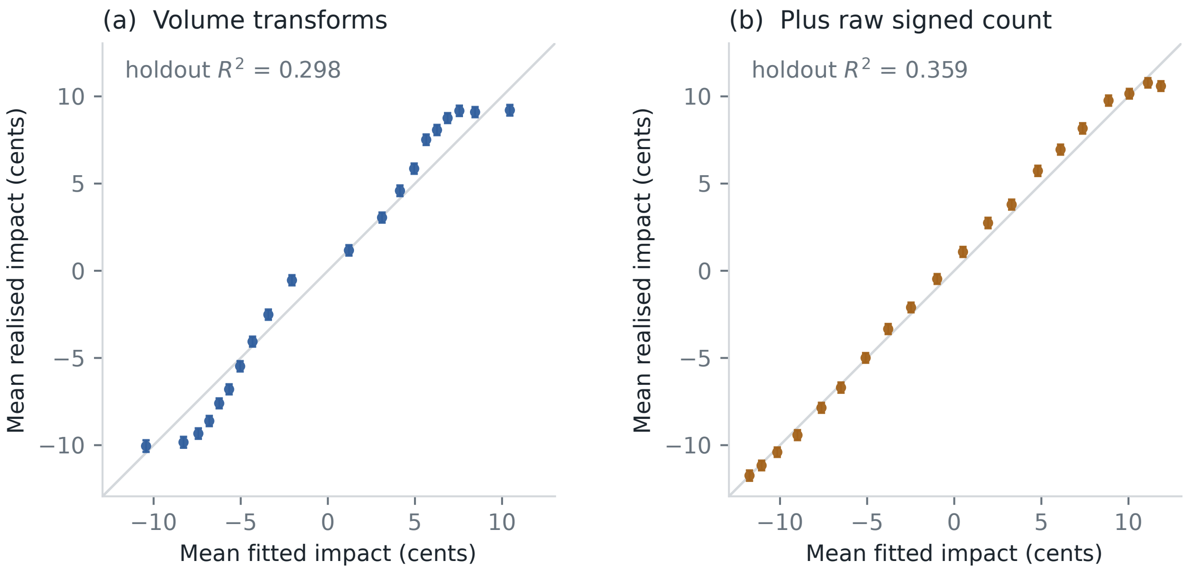

Figure 2. Holdout calibration by fitted-value quantile. Points show mean fitted and realized impact in 20 equally populated bins. Error bars are 1.96 observation-level standard errors and are included only as a visual guide; the formal score interval is clustered by date.

Figure 3 groups the holdout residual of the volume model by signed count. The mean moves from roughly −4 cents for ten sells to more than 3 cents for strong buy counts. This monotone residual pattern is the signal captured by ΔE.

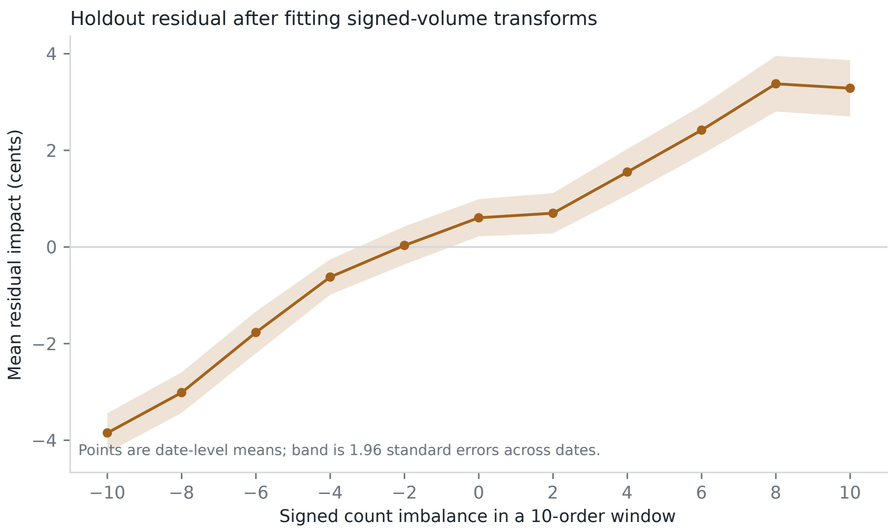

Figure 3. Holdout residual from the volume-transform model, grouped by signed order count. Points are equal-weighted means across dates. The band is 1.96 standard errors of the date-level means.

Adding raw count lowers MSE from 117.16 to 106.93 cents squared, a relative reduction of 8.7%. The day-clustered 95 percent interval is [8.0%, 9.5%]. The corresponding increase in R^2 is 0.061, with interval [0.056, 0.066]. None of the 10,000 bootstrap draws gives a non-positive improvement. This interval quantifies sampling variation across the 51 late-year dates. It does not cover model-selection or cross-year uncertainty. For context, the full count-transform model lowers MSE by 27.4% relative to linear signed volume; that is a total ablation gap, not the effect of count alone.

6.2 Time and horizon checks

The raw-count specification outperforms the nonlinear volume model in every expanding test block in Figure 4. Its R^2 ranges from about 0.34 to 0.38, while the volume-only score ranges from about 0.29 to 0.31. The gap is present before the final December block.

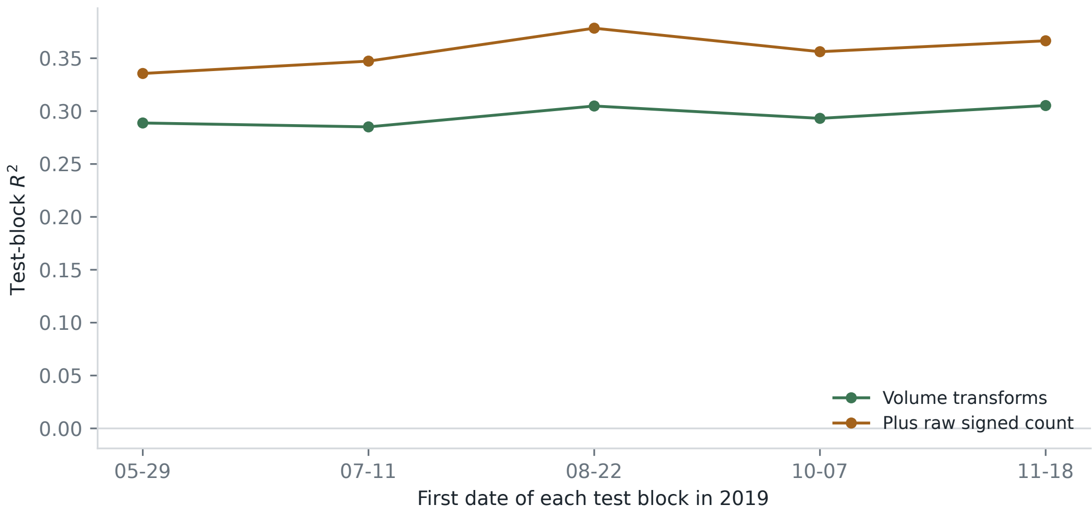

Figure 4. Five expanding-window evaluations. Each point is scored on the contiguous block beginning at the date shown. Earlier test blocks are added to the training set before the next fit.

The same ordering holds from 5 to 100 orders (Figure 5). At $T = 5$, R^2 rises from 0.237 to 0.282. At $T = 100$, it rises from 0.375 to 0.483. Longer windows average more idiosyncratic trade-level noise and contain a wider count range, so higher scores at larger T should not be read as better short-horizon forecasts.

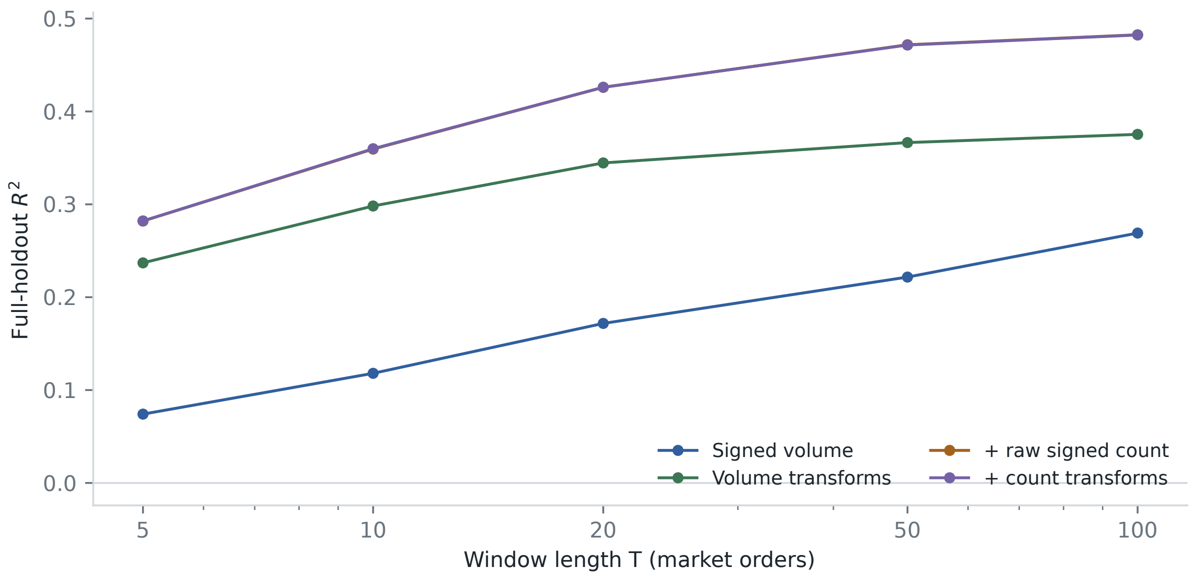

Figure 5. Full-holdout R^2 across event-time horizons. The raw count term accounts for nearly all of the gap between the volume-transform and full specifications.

The holdout result is also present within each calendar month. The augmented R^2 is 0.329 in the October portion, 0.378 in November, and 0.363 in December. With log return in basis points as the response, the scores are 0.281 and 0.307, and MSE falls by 3.7%. Price normalization reduces the size of the gain but does not remove it.

6.3 Why the previous score was 0.24 to 0.38

The earlier course protocol removed observations beyond three standard deviations in both signed volume and realized impact. Re-estimating those bounds on training dates avoids using the test distribution to set the cutoff, but applying the impact cutoff to the test still selects observations by the realized response. Under that trimmed comparison, the baseline has $R^2 = 0.241$ and the augmented model has $R^2 = 0.383$. MSE falls by 18.7%.

The trimmed scores describe the retained central sample. Selection on realized test impact makes them unsuitable as the primary evaluation. The cutoff raises the baseline much more than the count model, explaining the difference between 0.24 to 0.38 and the untrimmed 0.12 to 0.36 comparison.

7 Depth-aware L2 replay and next-price direction

7.1 Finite-depth transition audit

A finite-depth LOBSTER file is not a historical level-3 feed. Its message rows describe events that update the requested price range (LOBSTER, 2026). When a displayed level is depleted, a new tail level can enter from outside depth 2. Orders that formed that tail while it was unobserved are absent, so reconstructing their FIFO queues would invent state.

The C++20 audit seeds each included session from its first authoritative snapshot. It applies each later observable event to the previous two levels and compares the result with the supplied post-event snapshot. A transition is *exact* when every resulting level was known. It is *depth-censored* when depletion exposes an unobserved tail: the surviving prefix is checked and replay then resumes from the authoritative snapshot. Prices, sizes, and timestamps remain fixed-point integers.

Table 4. Included-session transition audit. The first row of each of the 249 sessions seeds replay and is not classified as a transition.

Quantity	Value
Message/snapshot pairs	38,516,432
Fully determined transitions	28,544,808 (74.11%)
Depth-censored transitions	9,971,375 (25.89%)
Observable mismatches	0
Unsupported events / invalid snapshots	0 / 0

Zero mismatches means that all observable prices and sizes agree under this contract. It does not turn level-2 data into an exchange matching engine, validate an unobserved queue, or establish execution priority.

7.2 Benchmark protocol

The benchmark separates complete CSV decoding from replay over the resident 64-byte event records. The measured replay audits transitions, checks snapshot invariants, and maintains a checksum. The aggregate signal exports below run separately after those measured passes.

Table 5. Single-threaded included-session benchmark on an Apple M4 Pro, 48 GB, compiled for arm64 with Apple Clang 17 in Release mode.

Phase	Run-average ns/event	Million events/s
CSV decode, median of 3 passes	115.19	8.68
Resident replay, median of 11 passes	9.22	108.46

Two unmeasured replay warm-ups precede the eleven passes. Their p95 run-average is 9.51 ns/event. The operating-system page cache is not flushed between decode passes: the first recorded pass costs 277.33 ns/event. Each timing divides a complete pass by 38,516,432. These are throughput measurements, not individual-event tail latency or wire-to-wire trading latency.

7.3 Causal queue and order-flow signals

For the displayed best queues q_t^b and q_t^a, define

$$I_t = \frac{q_t^b - q_t^a}{q_t^b + q_t^a}. \tag{6}$$

I also calculate the top-of-book OFI increment of Cont et al. (2014),

$$\begin{aligned} e_t = {} & \mathbf{1}_{\{P_t^b \geq P_{t-1}^b\}} q_t^b - \mathbf{1}_{\{P_t^b \leq P_{t-1}^b\}} q_{t-1}^b \\ & - \mathbf{1}_{\{P_t^a \leq P_{t-1}^a\}} q_t^a + \mathbf{1}_{\{P_t^a \geq P_{t-1}^a\}} q_{t-1}^a. \end{aligned} \tag{7}$$

Let $\tau(t)$ be the state at which the current mid-price was established. Only already observed increments enter

$$E_t = \sum_{u=\tau(t)+1}^{t} e_u, \qquad Z_t = \frac{2}{\pi} \arctan\left(\frac{E_t}{q_t^b + q_t^a}\right). \tag{8}$$

Thus E_t resets after each mid-price change, while Z_t is a bounded, depth-normalized summary of submissions, cancellations, and executions since that change. It is not a reconstructed queue or an estimate of order priority.

The label is one when the next same-session mid-price change is upward and zero when it is downward. I retain states after an event changes a best price or size. C++ aggregates (I_t, Z_t) onto a fixed 31×31 grid by date and spread regime; no row-level state enters the statistical package.

The first 198 included dates through 17 October fit four logistic specifications: intercept, I_t, Z_t, and (I_t, Z_t). All 51 dates from 18 October onward are scored without refitting. The intercept-only benchmark uses the training up-move frequency. Brier improvements and intervals resample complete test dates.

Table 6. Linear next-move signal ablation on the fixed late-2019 test dates. Brier reduction is relative to the training-frequency benchmark.

Spread	Signal	Observations	ROC-AUC	Brier reduction
All	Queue	5,489,604	0.538	0.12%
All	OFI	5,489,604	0.616	5.61%
All	Queue + OFI	5,489,604	0.627	5.61%
One tick	Queue	61,829	0.627	4.70%
One tick	OFI	61,829	0.694	12.06%
One tick	Queue + OFI	61,829	0.752	18.24%

Across all spreads, OFI raises ROC-AUC from 0.538 to 0.616 and reduces Brier score by 5.61%. The combined interval is [5.36%, 5.84%]. Queue adds no detectable pooled gain over OFI: the point improvement is 0.004 percent and its complete-date interval is [−0.09%, 0.10%]. In the exploratory one-tick subset, the combined model reaches 0.752 and a 18.24% reduction [16.33%, 19.97%]. Relative to OFI alone, the reduction is 7.02% [5.22%, 8.79%].

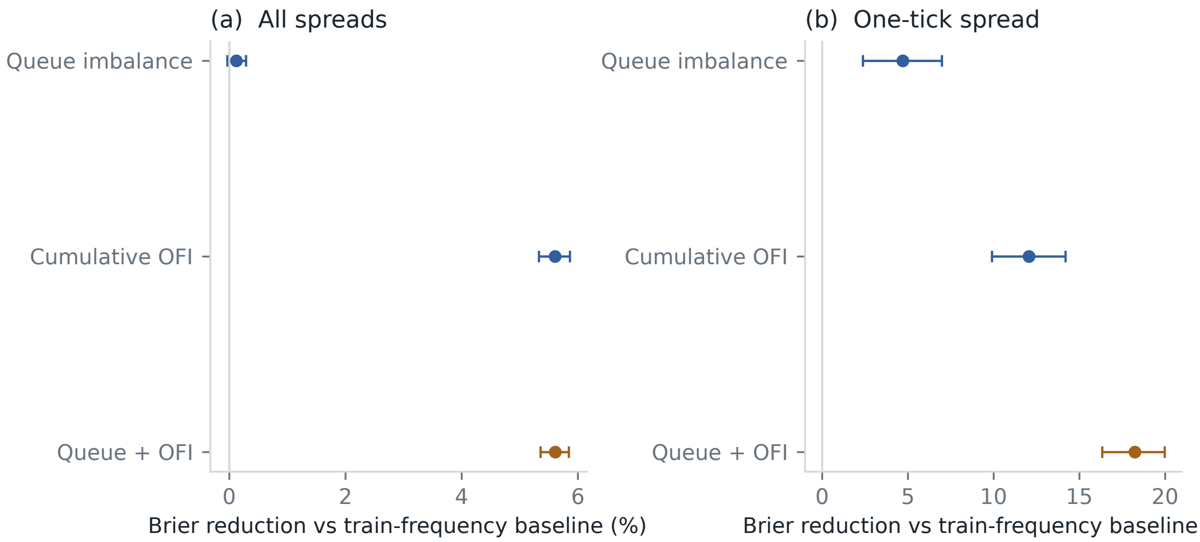

Figure 6. Relative Brier reduction against the training-frequency benchmark. Bars are 95 percent intervals from 10,000 resamples of the 51 complete test dates.

Repeating the complete ablation on fixed grids with 21, 31, and 41 bins per axis leaves the pooled combined ROC-AUC between 0.626 and 0.627 and its Brier reduction between 5.57 and

5.63 percent. The one-tick values remain 0.751–0.752 and 18.10–18.24 percent. The queue-only calibration below remains useful as a univariate diagnostic. A direction score before fees, latency, and queue position is not a trading strategy.

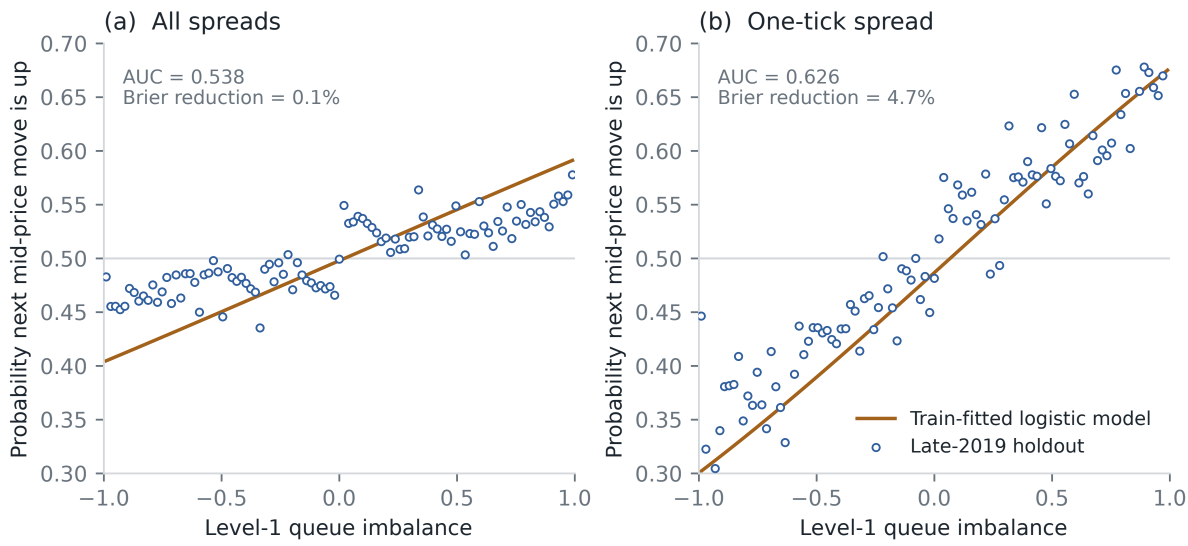

Figure 7. Next-move calibration for best-quote states. The orange curve is fitted on the first 198 included dates; blue circles are empirical frequencies on the final 51 dates. Test bins with fewer than 100 observations are omitted from the scatter but remain in the scores.

8 Scaling analysis

8.1 Construction

The scaling exercise follows the aggregate-volume form in Patzelt and Bouchaud (2018). For a window of N timestamp-aggregated type-4 and type-5 transactions, let Q be signed share volume after a daily-volume adjustment, and let r be the log mid-price change. Conditional curves are estimated in 31 quantile bins for 28 log-spaced horizons from 10 to 5,000 transactions.

Each curve is modeled as

$$\mathcal{R}_N(Q) = R_N F(Q/Q_N), \tag{9}$$

$$F(x) = \frac{x}{(1+|x|^{\alpha})^{\beta/\alpha}}. \tag{10}$$

One width Q_N and height R_N are estimated per horizon, with a common α and β. The nonlinear fit uses observation-count weights and a soft-L1 loss. Figure 8 plots the resulting rescaled curves. The collapse is close over the central observed range, with more dispersion around the origin and at a few long-horizon curves.

8.2 Scale fits and interpretation

After the curve fit, the scale parameters are regressed on horizon in log space:

$$Q_N = a_Q N^{\xi}, \qquad R_N = a_R N^{\psi}. \tag{11}$$

Table 7 and Figure 9 report those second-stage fits. The width exponent is 0.976 and the height exponent is 0.656. The residual panels show that the largest horizons carry the main deviations from a straight power law.

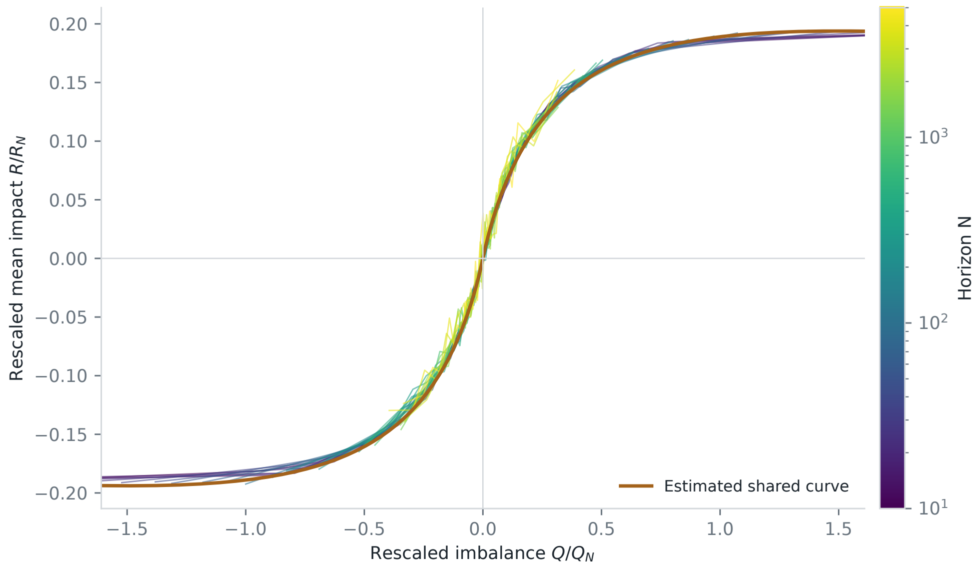

Figure 8. Finite-size scaling collapse for TSLA 2019. Color identifies the aggregation horizon. The orange line is the common fitted shape. This is an in-sample curve construction, not a forecast evaluation.

Table 7. Log-log fits to the 28 estimated scale parameters. Standard errors condition on the first-stage scale estimates.

Scale	Exponent	Standard error	In-sample R^2
Width Q_N	0.976	0.012	0.9958
Height R_N	0.656	0.006	0.9975

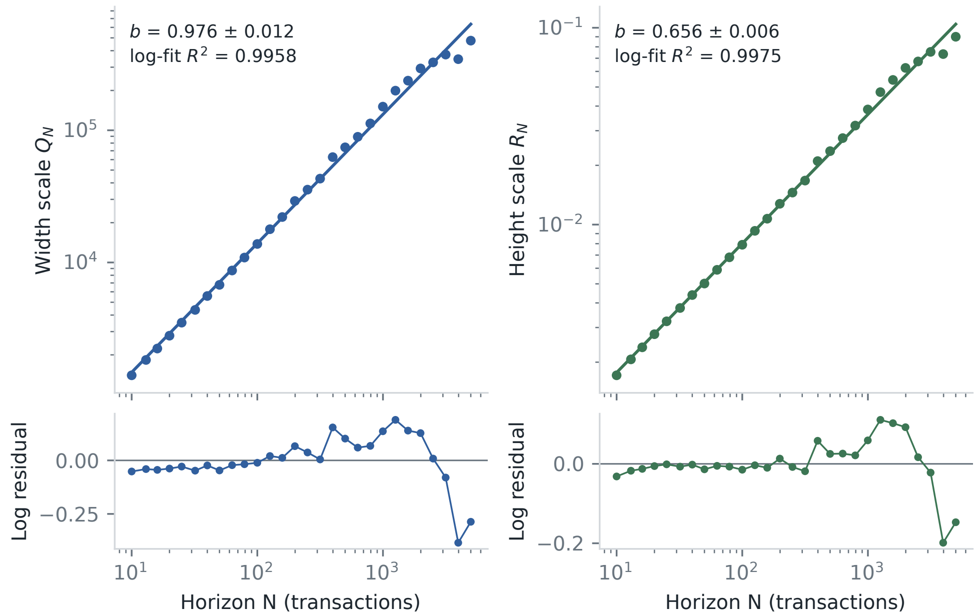

Figure 9. Power-law fits for the estimated width and height scales. Lower panels show log residuals. The displayed R^2 values describe these 28 in-sample points only.

Near zero imbalance, the master-curve slope scales as R_N/Q_N. The fitted exponents imply a decay $N^{-(\xi-\psi)}$ with $\xi - \psi = 0.320$. The local Kyle slopes in Section 4 decay with exponent

0.315. The two calculations use different transaction samples and return definitions, so their agreement is descriptive.

The pooled order-sign standard deviation scales with horizon exponent 0.693. This is a descriptive variance-scaling fit, not a DFA estimate or sufficient evidence of long memory. The 0.9958 and 0.9975 values belong to regressions on 28 estimated scale points. They do not measure future price variation. The proposed scaling form describes this included TSLA sample in-sample; the cross-asset claim in Patzelt and Bouchaud (2018) requires a broader panel.

9 Limitations

- The variables and response cover the same window, and order flow is endogenous. The fitted coefficients are neither pre-trade forecasts nor causal price effects.
- The replay checks finite-depth state transitions against authoritative snapshots. It cannot recover orders submitted outside the requested depth, FIFO priority, network timing, or exchange matching logic.
- Timestamp and side grouping cannot recover investor identity or institutional metaorders. The book covers NASDAQ rather than consolidated US liquidity, and hidden-execution signs are uncertain around the midpoint.
- Several order-book states can share the same eventual mid-price-change label. The scores do not represent independent event trials; uncertainty is resampled at the trading-date level. OFI also depends on the waiting horizon to that next move, so another horizon defines a different feature.
- The one-tick spread stratification is exploratory. Its stronger result is not evidence of generalization beyond this stock-year.
- The joint logistic models use fixed binned features rather than raw event states. The 21/31/41-grid check limits this discretization concern but does not replace validation on an untouched stock-year.
- The feature family grew out of the course analysis. Although dates are held out chronologically, there is no second untouched year after the feature comparison. The bootstrap does not account for that research process.
- Three delivered source files end before their scheduled close, and replacement files are unavailable under the original project access. The canonical rebuild declares them as source exclusions and uses the other 249 sessions. This source limitation prevents a claim of complete 2019 coverage. The calendar boundaries are fixed rather than recomputed after exclusion.
- The scaling standard errors treat first-stage Q_N and R_N estimates as data. They do not propagate all curve-fitting uncertainty, and overlapping information across horizons remains.
- TSLA in 2019 is one liquidity regime. The count, OFI, and fitted-scaling results cannot be assumed to hold for another asset or year.

10 Reproducibility and provenance

The Python package first loads the shared source and calendar policy and writes an aggregate scheduled-close audit. It then reconstructs the two transaction tables from the included sessions, builds event-time windows, fits the models, and writes the remaining aggregate files under `results/`. The C++20 library loads the same policy, rejects undeclared incomplete sessions, audits finite-depth transitions, and exports daily queue and joint queue/OFI bins. The figures are vector PDFs. Synthetic tests cover file pairing, source coverage, half-day filtering, book alignment, day boundaries, fixed chronological splits, complete-date resampling, queue/OFI ablation, grid robustness, and the scaling regressions. Reproducing the annual transaction tables, replay, and aggregate grids requires licensed LOBSTER files, which are not tracked.

The course presentation lists L. Cohen, R. Darwish, A. Pariente, H. Rohrbach, and R. Sithisak. Anastasia Bugaenko supervised the work. I wrote this standalone package and report after the group project; the collaborative repository remains separate.

11 Conclusion

On the included-session late-2019 holdout, adding raw signed count to the nonlinear volume model moves R^2 from 0.298 to 0.359 and reduces MSE by 8.7%, with a date-bootstrap interval of [8.0%, 9.5%]. The full count-transform model's 27.4% reduction relative to linear signed volume is a broader ablation gap and is not attributed to count alone. The incremental ordering is stable across the time and horizon checks reported above.

The finite-depth engine audits 38,516,432 included events without claiming unobservable FIFO state and replays the resident sample at 108.46 million events/s on the stated machine. Causal OFI is more informative than the static queue state across all spreads: the combined model reaches ROC-AUC 0.627 and reduces Brier score by 5.61% on later dates. Queue and OFI complement each other in the exploratory one-tick subset, where the combined ROC-AUC is 0.752.

The included TSLA sessions also fit the scaling form proposed by Patzelt and Bouchaud (2018). This is a within-sample application to one stock-year, not an external replication. The supervised model is evaluated by the chronological score.

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